PROJECT DEVELOPMENT & IMPLEMENTATION REPORT

Technical Development Log, Setup Chronicles, and Refactoring Steps

1. Environment & Setup Chronicles

Development was initiated individually by setting up a Python virtual environment (`env`) under the project parent directory. Required libraries were installed via `pip` including PyTorch, Sentence-Transformers, Whisper, Librosa, Streamlit, and ReportLab. The code was structured modularly to isolate the UI, database connection, and individual intelligence modules.

2. Code Refactoring & Troubleshooting

During development, two major technical hurdles were resolved:

1. **The ffmpeg Dependency Issue**: Initial testing of Whisper threw a `FileNotFoundError` because Whisper's internal loader calls `ffmpeg` via a subprocess. We refactored `modules/transcription.py` to use `librosa.load(audio_path, sr=16000, mono=True)` which reads files natively, completely bypassing Whisper's ffmpeg dependency.

2. **ReportLab Unicode Rendering Crashes**: Emojis (like sparkles, thumbs-up, warning icons) used in feedback notes crashed ReportLab's standard Helvetica font. We implemented a custom `strip_emojis(text)` helper in `modules/report_generator.py` to clean all inputs before compiling the PDF.

3. Relational Schema Mapping

We constructed `database.py` utilizing SQLAlchemy's declarative base models. We wrote context-managed database sessions (`get_db()`) to prevent resource leaks and automatically seed default target concepts.